LMWiRe: an R package for Linear Modelling of Wide Responses based on ASCA family of methods.

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Abstract

Two recent multivariate methodologies, namely ASCA (ANOVA-Simultaneous Component Analysis) and APCA (ANOVA-Principal Component Analysis), have been developed to analyse multivariate data derived from an advanced experimental design. We present the R package `LMWiRe` [1] that implements a revised version of ASCA/APCA to deal with unbalanced designs. It comes with additional methodology improvement and functionalities.

ASCA+ methodology

Classical regression (e.g. MANOVA) cannot deal with multicollinear high dimensional experiment data (**Y**) having a low number of observations/high number of variables.

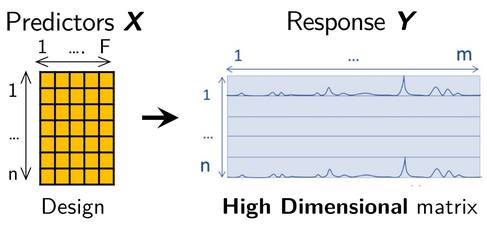

Figure 1: Multiple regression paradigm.

ASCA/APCA [2,3] were developed to analyse these data with fixed categorical predictors in a multivariate fashion by combining **modelling** (ANOVA) and **dimension reduction** (PCA). ASCA+/APCA+ [4] are extended versions for **unbalanced designs** based on the GLM theory.

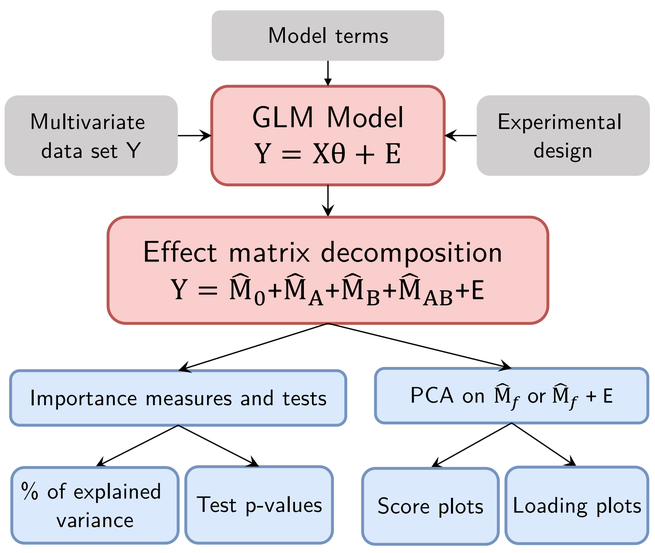

Figure 2: Summary of ASCA+/APCA+ methodology.

Step 1: **Parallel GLM** on each response and decomposition into **effect matrices**.

Step 2: Effect matrix interpretation:

- **Importance measure** based on type III Sum of Squares.
- **Significance tests** of the effects based on scalable parametric bootstrap.
- **PCA** on the effect matrices: pure (**ASCA**) or augmented by the residuals (before PCA: **APCA** ; after PCA: **ASCA-E**) to visualize the model results through informative loading and score plots. APCA/ASCA-E score plots include individual variations between observations not explained by the model to give indication on the effects significance.

LMWire package overview

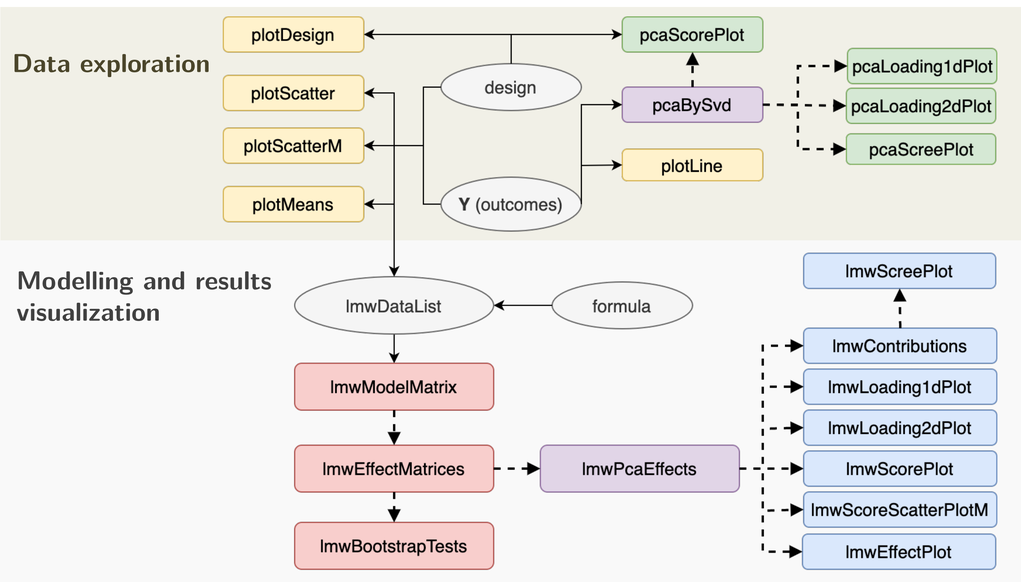

Figure 3: Representation of the `LMWiRe` set of functions. Rectangle: function; ellipse: data; plain arrow: the data is inputted in the targeted function; dashed arrow: the value of the output function is inputted in the next one.

Two sets of functions:

(1) Data exploration: Explore and visualise the data included in the outcomes and design. Perform PCA on the experiment data and visualise PCA results.

(2) Modelling and results visualisation: Parallel modelling to estimate the effect matrices and quantify their relative importance. Use bootstrap to test their significance. Apply PCA to the effect matrices. Visualise the Principal Components (PC) with scores, loadings and effects plots.

Case study presentation

The UCH dataset [5] included in `LMWiRe`, is referred as the `lmwDataList` list object with the following elements:

- `Outcome`: metabolomic spectral profiles obtained by ^{1}H NMR spectroscopy of samples from a pool of rat urine samples. The pool of samples were spiked with different known concentrations of Hippurate and Citrate at different Times and Days.

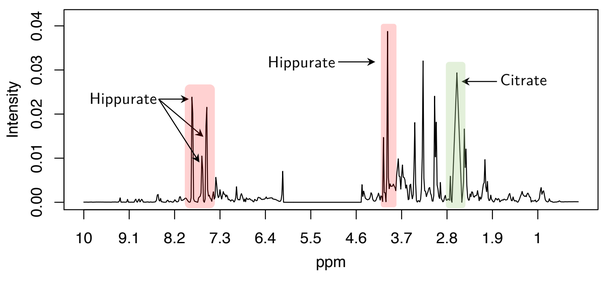

Figure 4: Example of an ^{1}H NMR spectral profile.

- `Design`: 4 factors (Hippurate, Citrate, Time and Day), slightly unbalanced design.
- `Formula`: Full three-way ANOVA model:

```
outcomes ~ Hippurate + Citrate + Time + Hippurate:Citrate + Time:Hippurate +
           Time:Citrate + Hippurate:Citrate:Time
```

Results

Visual representation of the data

The experimental design is slightly unbalanced due to outlier removal (Fig. 5). PCA on the outcomes does not clearly separate the 2 chemical products i.e. PCs are a mixture of the spectral profiles of Citrate and Hippurate (Fig. 6).

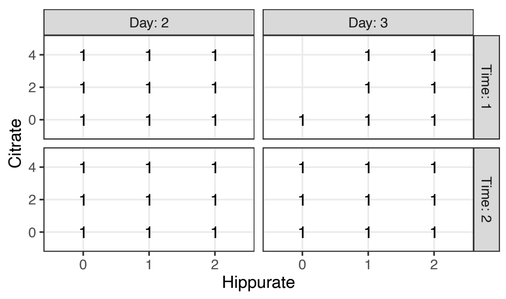

Figure 5: Design of the UCH dataset obtained with `plotDesign`.

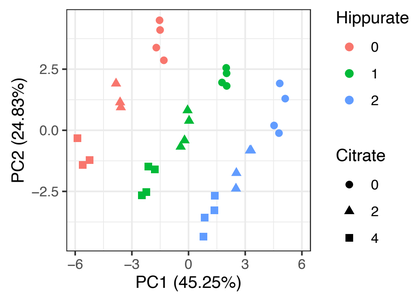

Figure 6: PCA score plot (PC1-PC2) of the outcome matrix (`pcaScorePlot`).

Effects importance quantification and significance test

The main effects `Hippurate`, `Citrate` and `Time` and the interaction `Hippurate:Time` are significant. Together these effects explain 91.69% of the variance of the outcomes (Table 1).

	Observed p-value	% of variance
Hippurate	< 0.001	39.31
Citrate	< 0.001	29.91
Time	< 0.001	16.24
Hippurate:Citrate	0.147	1.54
Hippurate:Time	< 0.001	6.23
Citrate:Time	0.418	0.54
Hippurate:Citrate:Time	0.089	1.68
Residuals		4.3

Table 1: Bootstrap test p-values (1000 resampling events) and percentages of explained variance for each model effect (`lmwBootstrapTests`).

Effects visualisation

Score plot for the main effect of Hippurate effect (Fig. 7) shows a clear separation between the different concentration levels, suggesting a strong effect, confirmed by the significance test. The associated loading clearly retrieves the Hippurate peaks.

Analysis of the residuals indicate a remaining but limited effect of Day. Not displayed here, a score scatter plot of the significant interaction term (`Hippurate:Time`) is less straightforward to interpret.

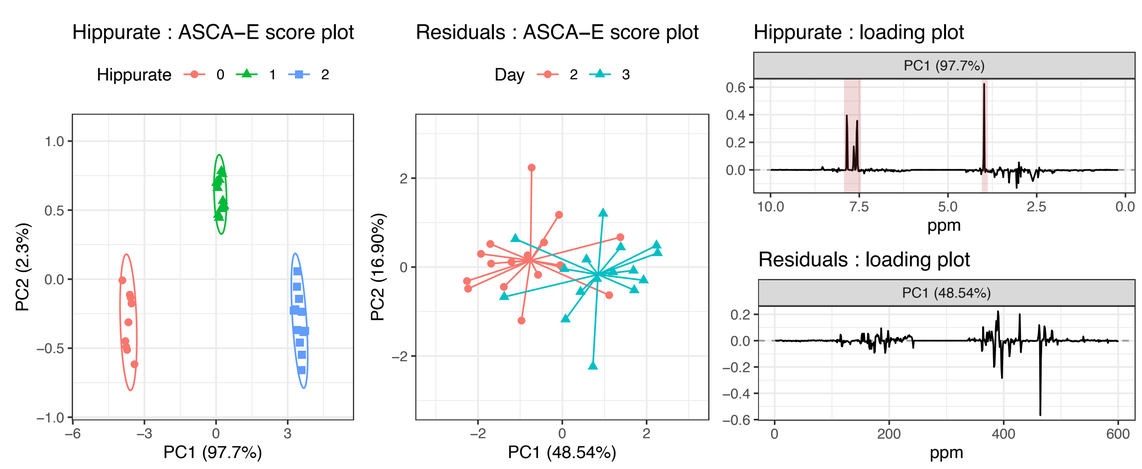

Figure 7: PCA Scores and loadings of APCA for the Hippurate effect and the model residuals (`lmwScorePlot` and `lmwLoading1dPlot`).

Instead, the ASCA scores for this interaction effect or for a combination of effects can be represented as a line plot (Fig. 8) according to the levels of one main effect. When displayed alone, the `Hippurate:Time` interaction effect seems important but is rather small once combined with the main effects of `Hippurate` and `Time`.

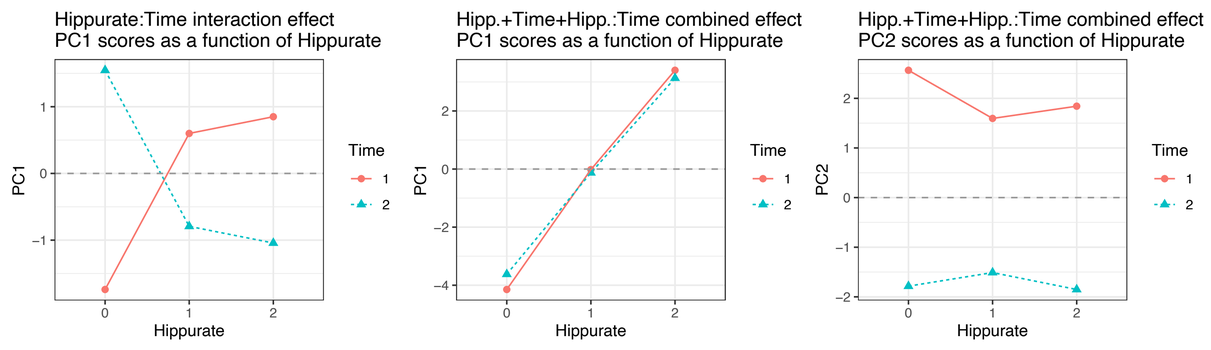

Figure 8: Effect plots (`lmwPcaEffects`) for the interaction term (`Hippurate:Time`) and a combination of effects (`Hippurate + Time + Hippurate:Time`).

Conclusions

`LMWiRe` applies ASCA+ methodology to any (un)balanced experimental design with fixed categorical factors. It offers optimized methods to calculate effect importance and test their significance. It comes with a comprehensive results interpretation toolkit where the data and ASCA/APCA results are represented with customizable ggplot2-based graphical outputs. The package is open to future extensions such as more sophisticated statistical models, e.g. linear mixed models as in [6].

References

[1] Thiel, M., Benaiche, N., Martin, M., Franceschini, S., Van Oirbeek, R., Govaerts, B. LMWiRe: an R package for Linear Modeling of Wide Responses based on ASCA family of methods. *submitted*

[2] Smilde, A. K., Jansen, J. J., Hoefsloot, H. C. J., Lamers, R.-J. A. N., van der Greef, J., Timmerman, M. E. (2005). Anova-simultaneous component analysis (ASCA): a new tool for analyzing designed metabolomics data, *Bioinformatics 21* (13) 3043–3048.

[3] Harrington, P., Vieira, N. E., Espinoza, J., Nien, J. K., Romero, R., Yergey, A. L. (2005). Analysis of variance–principal component analysis: A soft tool for proteomic discovery, *Analytica chimica acta 544* (1) 118–127.

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[6] Martin, M., & Govaerts, B. (2020). LiMM-PCA: Combining ASCA+ and linear mixed models to analyse high-dimensional designed data. Journal of Chemometrics, 34(6), e3232.

https://github.com/ManonMartin/LMWiRe